SMAIN FEDI

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PROFILE

Cybersecurity Engineering Student at EPI Digital School, with a Bachelor's background in Computer Systems and Networks Engineering. Hands-on experience in network security, system administration, and intrusion detection through academic projects. Interested in defensive cybersecurity, security monitoring, and incident investigation.

EDUCATION

Engineering Degree in Cybersecurity 2026 – Present
EPI Digital School – Sousse, Tunisia

Bachelor's Degree in Computer Systems and Networks Engineering 2022 – 2026
University of Gabès – Faculty of Sciences

Baccalaureate in Mathematics 2021 – 2022
Malloulech High School

TECHNICAL SKILLS

Networking: TCP/IP, IPv4/IPv6, subnetting, VLANs, ACLs, DHCP, DNS, network segmentation (WAN, DMZ, LAN, MGMT), pfSense firewall, Cisco Packet Tracer.

Systems: Linux (Ubuntu, Kali), Windows / Windows Server, user and permission management, basic hardening, VirtualBox labs.

Cybersecurity: Signature-based and behavioral intrusion detection, network traffic analysis, attack simulation, security monitoring.

Machine Learning & Programming: Python, Scapy, Random Forest, feature extraction, Flask, C, Java, SQL, PL/SQL.

Tools: Wireshark, Snort, pfSense, Nmap, SlowHTTPTest, Scapy, Packet Tracer, VirtualBox.

PROJECTS

Hybrid Behavioral Intrusion Detection System (Cybersecurity PFE)
Designed and implemented a hybrid intrusion detection architecture combining pfSense, Snort, and a Random Forest-based behavioral detection pipeline in a VirtualBox environment.

- Deployed a segmented virtual network with five VMs across WAN, DMZ, LAN, and MGMT security zones.
- Developed a Machine Learning-based IDS trained on 200,000 samples, extracting 71 network traffic features using Scapy.
- Generated and analyzed attack traffic (Nmap scans, brute-force attempts, Slow HTTP attacks, DoS scenarios) to evaluate detection capabilities.
- Compared Snort signature-based detection with behavioral AI detection, showing improved identification of attack patterns, including source IP changes.
- Implemented a real-time traffic classification pipeline with confidence scores and security monitoring.

Technologies: pfSense, Snort, Python, Scapy, Random Forest, VirtualBox, Nmap, SlowHTTPTest, Flask.

Smart IoT Parking System: Developed an IoT-based parking management prototype with sensors, backend logic, a web dashboard, and real-time parking reservation functionality.

CarFlow – Car Rental Web Application: Served as Product Owner in a Scrum-based team project, contributing to project planning, application architecture, backlog management, and database coordination.

CERTIFICATIONS

Cisco Networking Academy: Introduction to Cybersecurity, Ethical Hacker, Apply AI: Analyze Customer Reviews.

Fortinet: Hands-On Defense with Fortinet Security Fabric. **DataCamp:** Introduction to SQL, Intermediate SQL.

OPSWAT Academy: Introduction to Critical Infrastructure Protection.

LANGUAGES

Arabic: Native — French: Fluent — English: B2 (IELTS Academic, 6.0)